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# Lecture 6: Entropic OT, continued

## 1 Review: Intro to Entropic OT

### 1.1 Kantorovich's Problem

Consider two probability measures $\alpha \in \mathcal{P}(X)$ and $\beta \in \mathcal{P}(Y)$ on measurable spaces $X$ and $Y$, and a cost function $c : X \times Y \to \mathbb{R}$. Kantorovich's problem seeks an optimal transport plan $\pi \in \Pi(\alpha, \beta)$ minimizing:

$$\inf_{\pi \in \Pi(\alpha,\beta)} \int_{X \times Y} c(x, y)\, d\pi(x, y),$$

where $\Pi(\alpha, \beta)$ denotes the set of couplings with marginals $\alpha$ and $\beta$.

The **dual formulation** is given by:

$$\inf_{f,g: f(x)+g(y) \le c(x,y)} \left( \int_X f\, d\alpha + \int_Y g\, d\beta \right).$$

Here, the constraint $f(x) + g(y) \le c(x, y)$ captures the implicit interaction between potentials $f$ and $g$. To make this interdependence of $f$ and $g$ explicit, in the last lecture, we introduced the $c$-**transform** and $c$-conjugacy:

$$f^c(y) := \inf_{x \in X} \big( c(x, y) - f(x) \big).$$

This allows us to reduce the dual formulation to a single function variable, yielding:

$$\sup_{f \in \text{c-CVX}} \left( \int_X f(x)\, d\alpha(x) + \int_Y f^c(y)\, d\beta(y) \right).$$

This representation highlights the geometry induced by the cost function $c$ and connects the problem to convex analysis.

### 1.2 Wasserstein Distances as Special Cases

The Kantorovich duality recovers well-known metrics on probability measures:

- **1-Wasserstein distance:**
$$W_1(\alpha, \beta) = \sup_{\|f\|_{\text{Lip}} \le 1} \left( \int f\, d\alpha - \int f\, d\beta \right).$$
- **2-Wasserstein distance (squared):**
$$W_2^2(\alpha, \beta) = \sup_{\varphi \text{ convex}} \left( \int \varphi(x)\, d\alpha(x) + \int \varphi^*(y)\, d\beta(y) \right),$$
where $\varphi^*$ denotes the convex conjugate of $\varphi$.

### 1.3 Discrete Version and Entropy Regularization

In the discrete setting, with $\alpha = a \in \Delta^n$ and $\beta = b \in \Delta^m$, and cost matrix $C \in \mathbb{R}^{n \times m}$, the Kantorovich problem becomes:

$$\min_{\pi \in \Pi(a,b)} \langle \pi, C \rangle,$$

where $\langle \pi, C \rangle = \sum_{i,j} \pi_{ij} C_{ij}$.

To improve computational tractability (among other nice properties), we can add an **entropy regularization term**:

$$\min_{\pi \in \Pi(a,b)} \left( \langle \pi, C \rangle - \varepsilon H(\pi) \right),$$

where

$$H(\pi) = -\sum_{i,j} \pi_{ij} \log \pi_{ij}$$

is the entropy, and $\varepsilon > 0$ controls the regularization strength.

- For $\varepsilon \to 0$: the optimal plan $\pi$ concentrates on sparse couplings, approximating the original OT solution.
- For $\varepsilon \to \infty$: $\pi$ approaches the independent coupling $a \otimes b$ (maximum entropy solution).
- Intermediate $\varepsilon$: balances sparsity and smoothness, leading to numerically stable and efficient approximations.

# 2 Sinkhorn

## 2.1 The Algorithm

Recall that the entropy-regularized problem admits an efficient iterative scaling solution known as the **Sinkhorn algorithm**. Define:

$$K = \exp\left(-\frac{C}{\varepsilon}\right),$$

and initialize scaling vectors $u, v \in \mathbb{R}_+^n, \mathbb{R}_+^m$. The algorithm then proceeds as follows:

1. Initialize $u^{(0)} = \mathbf{1}$.
2. Iterate until convergence:

$$u \leftarrow \frac{a}{Kv}, \qquad v \leftarrow \frac{b}{K^\top u}.$$

The optimal transport plan is then recovered as:

$$\pi^\star = \operatorname{diag}(u)\, K \operatorname{diag}(v).$$

This algorithm scales well to large datasets and has become a cornerstone of computational OT.

## 2.2 Sinkhorn Complexity and Convergence

Each iteration involves:

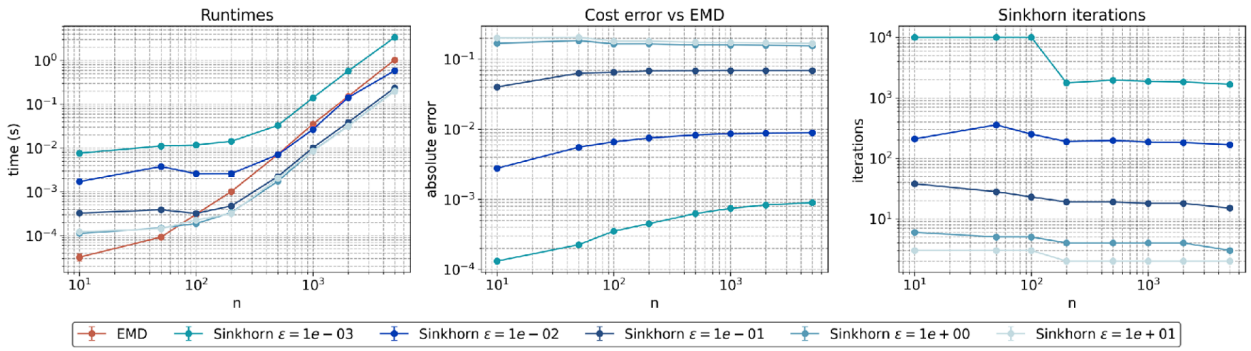
- Two matrix-vector multiplications: $Kv$ and $K^\top u$.
- Two elementwise vector divisions: $a \oslash (Kv)$ and $b \oslash (K^\top u)$.

The dominant cost is the matrix-vector multiplication, requiring $O(n^2)$ operations. Thus, each iteration of Sinkhorn has complexity $O(n^2)$.

Empirically, Sinkhorn iterations converge in approximately $O\left(\frac{1}{\varepsilon^2}\right)$ iterations to reach a prescribed accuracy (though precise theoretical rates are still an active area of research. This far exceeds the rate of LP solvers.

## 2.3 Comparison

Sinkhorn is often *orders of magnitude faster* than LP solvers for large-scale problems. In addition, with entropy regularization, convergence is fast in practice, especially for moderate $\varepsilon$. Since $K = e^{-C/\varepsilon}$, choosing a small $\varepsilon \ll 1$ (i.e., small entropy term) produces $\lambda(K)$ close to 1 and results in convergence being slow to compute. Conversely, increasing the entropy results in a subsequent increase in the convergence rate. Studying the convergence properties of the Sinkhorn algorithm is a very active area of research!




In addition, we see from the plot above that while using EMD (Earth Mover's Distance) may be faster for some small regime of $n$, Sinkhorn begins to take over as $n$ increases. From the error graph, we see that increased regularization in the Sinkhorn algorithm trades off a higher cost error than EMD for faster convergence.

# 3 Convergence Results

## 3.1 Hilbert's Projective Metric

Convergence analysis is greatly simplified using *Hilbert's projective metric*, which is scale-invariant and thus well-suited for Sinkhorn, since the pair $(u, v)$ is only defined up to a multiplicative constant.

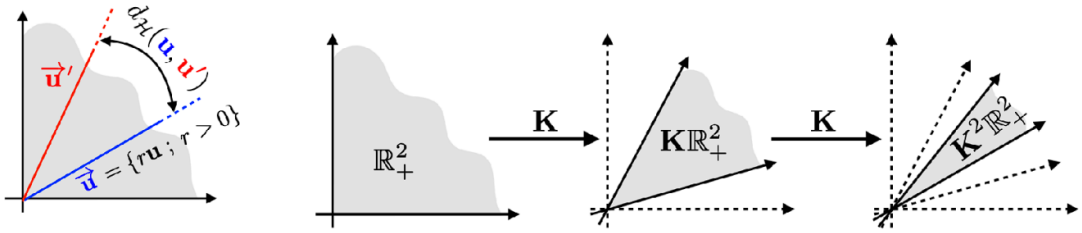



**Figure 4.7:** Left: the Hilbert metric $d_{\mathcal{H}}$ is a distance over rays in cones (here positive vectors). Right: visualization of the contraction induced by the iteration of a positive matrix $\mathbf{K}$.

Key properties:

- **Guaranteed convergence:** Sinkhorn iterates $(u^{(t)}, v^{(t)})$ converge to the entropic OT solution.
- **Geometric rate:** Convergence is geometric in Hilbert's metric, governed by $\lambda(K)$.
- **Contraction:** $\lambda(K) < 1$, ensuring contraction in each iteration.

The contraction coefficient $\lambda(K)$ measures the *spread or imbalance* of values in the kernel $K$.

- If rows of $K$ are *balanced* (entries of similar magnitude), then $\eta(K) \gtrsim 1$ and thus $\lambda(K)$ is small, implying fast convergence since $O(\lambda(K)^{2l})$ decays quickly.
- If rows of $K$ are *uneven/"spikey"* (large dynamic range across entries), then $\eta(K)$ is large and $\lambda(K) \lesssim 1$, implying slow convergence since $O(\lambda(K)^{2l})$ decays slowly.

## 3.2 Global Convergence Theorem

Let $K = \exp(-C/\varepsilon)$ denote the Gibbs kernel, and $(u^{(t)}, v^{(t)})$ the scaling vectors produced by Sinkhorn's iterations at step $t$.

**Theorem 1** (Sinkhorn Global Convergence)**.** *The iterates $(u^{(t)}, v^{(t)})$ converge to the unique scaling vectors $(u^*, v^*)$ associated with the entropic optimal transport solution:*

$$(u^{(t)}, v^{(t)}) \to (u^*, v^*).$$

*Moreover, in Hilbert's projective metric $d_{\mathcal{H}}$, one has:*

$$d_{\mathcal{H}}(u^{(t)}, u^*) = O\big(\lambda(K)^{2t}\big), \quad d_{\mathcal{H}}(v^{(t)}, v^*) = O\big(\lambda(K)^{2t}\big),$$

*where the contraction coefficient $\lambda(K)$ is given by*

$$\lambda(K) = \frac{\sqrt{\eta(K)} - 1}{\sqrt{\eta(K)} + 1}, \quad \eta(K) = \max_{i,j,k,\ell} \frac{K_{i,k} K_{j,\ell}}{K_{i,\ell} K_{j,k}}.$$

*Proof.* In Hilbert's metric

$$d_{\mathcal{H}}(x, y) = \log\left(\frac{\max_i x_i / y_i}{\min_i x_i / y_i}\right),$$

inversion and diagonal scalings are isometries, while Birkhoff's theorem gives

$$d_{\mathcal{H}}(Kx, Ky) \le \lambda(K)\, d_{\mathcal{H}}(x, y).$$

Thus the Sinkhorn map $\Phi(v) = b/(K^\top(a/(Kv)))$ is a $\lambda(K)^2$-contraction.

By Banach's fixed point theorem, $(u^{(\ell)}, v^{(\ell)}) \to (u^*, v^*)$ with

$$d_{\mathcal{H}}(u^{(\ell)}, u^*),\ d_{\mathcal{H}}(v^{(\ell)}, v^*) = O(\lambda(K)^{2\ell}).$$

Moreover, with $P^{(\ell)} = \operatorname{diag}(u^{(\ell)}) K \operatorname{diag}(v^{(\ell)})$,

$$d_{\mathcal{H}}(u^{(\ell)}, u^*) \le \tfrac{d_{\mathcal{H}}(P^{(\ell)} 1_m, a)}{1-\lambda(K)^2}, \quad d_{\mathcal{H}}(v^{(\ell)}, v^*) \le \tfrac{d_{\mathcal{H}}(P^{(\ell)\top} 1_n, b)}{1-\lambda(K)^2},$$

and

$$\| \log P^{(\ell)} - \log P^* \|_\infty \le d_{\mathcal{H}}(u^{(\ell)}, u^*) + d_{\mathcal{H}}(v^{(\ell)}, v^*).$$

Hence Sinkhorn converges geometrically with rate $\lambda(K)^{2\ell}$.

See Textbook, Theorem 4.2 (page 71) for reference [PC19]. □

## 3.3 Effect of $\varepsilon$

When $\varepsilon$ is small, the kernel $K = e^{-C/\varepsilon}$ becomes highly peaked and uneven:

- Entries of $K$ concentrate on small subsets of indices.
- $\eta(K)$ increases, worsening $\lambda(K)$.
- Convergence slows significantly, explaining the observed inefficiency of Sinkhorn for very small $\varepsilon$.

A smaller $\varepsilon$ couples $\pi_\varepsilon$ close to the true solution, but slower convergence can produce numerical instability (underflow/overflow) in finding $K = e^{-C/\varepsilon}$. In general, a larger $\varepsilon$ produces a smoother coupling and "blurrier" transport.

- We often treat $\varepsilon$ as a tunable hyperparameter in ML. One potential heuristic is to set $\varepsilon$ proportional to a typical transport cost (for example, the median of the entries of $C$).
- This can be chosen via cross-validation, where we can simplify our choice by taking normalized distances $C \leftarrow C/\text{mean}(C)$ and sweep $\varepsilon \in \{10^{-3}, \ldots, 10^{1}\}$.
- More generally, just like in ML, we observe the common theme of bias-variance trade-off via $\varepsilon$. A small $\varepsilon$ corresponds to high variance solutions that are accurate but unstable/slow, while a large $\varepsilon$ increases the bias via stable/fast but approximate solutions.

Sinkhorn convergence analysis continues to be refined, especially regarding small $\varepsilon$ regimes and nonstandard cost functions.

# 4 Barycentric Projection

## 4.1 From Couplings to Maps

The Kantorovich formulation of optimal transport yields a *coupling* $\pi \in \Pi(\alpha, \beta)$, i.e., a joint distribution over $(X, Y)$ with marginals $\alpha$ and $\beta$. Importantly, this provides a *plan*, not a deterministic map $T : X \to Y$.

- If $\pi$ happens to be a *permutation matrix*, then it corresponds directly to a Monge map:

$$\pi_{ij} \neq 0 \quad \Rightarrow \quad T(x_i) = y_j.$$

- For dense couplings, we cannot recover a unique map, but we can still define an *approximate* Monge map via the **barycentric projection**.

## 4.2 Definition

For empirical measures $\alpha = \sum_i a_i \delta_{x_i}$ and $\beta = \sum_j b_j \delta_{y_j}$ with coupling $\pi \in \mathbb{R}^{n \times m}$, the barycentric projection defines:

$$\tilde{T}(x_i) = \frac{1}{a_i} \sum_j \pi_{ij} y_j.$$

More generally, for arbitrary measures:

$$T(x) = \int_Y y \, \frac{d\pi(x, y)}{d\alpha(x)\, d\beta(y)} \, d\beta(y).$$

In essence, we express $x_i$ as a linear combination of the target points, where the weights are the coupling entries (in the discrete form), and so every source $x_i$ gets a map to a linear combination of the targets it transports to. Geometrically, the coupling can be imagined as defining some convex hull on the target $y_j$ vertices.

Given data matrices $X \in \mathbb{R}^{n \times d}$ and $Y \in \mathbb{R}^{m \times d}$, the barycentric projection of all points can be expressed compactly as:

$$\tilde{T}(X) = \mathrm{diag}(a)^{-1} \pi Y,$$

where $a = \pi \mathbf{1}_m$ are the marginals of $\pi$ with respect to $X$.

The barycentric projection can be understood as the conditional expectation:

$$\tilde{T}(x) = \mathbb{E}_\pi[Y \mid X = x].$$

Thus, $\tilde{T}$ maps each source point $x$ to the expected target location under the transport plan $\pi$.

## 4.3 Usefulness & Limitations

The barycentric projection yields an *approximate Monge map*, even when the original Kantorovich solution does not correspond to a true Monge solution. This approximation has important applications:

- Domain adaptation (e.g., mapping features between domains).
- Generative modeling (e.g., transporting samples between distributions).
- Style transfer and geometric transformations.

However, there are **limitations**,

- The barycentric projection is defined only on training/source samples $x_i$.
- There is no direct out-of-sample extension, unlike true Monge maps (since $\tilde{T}$ is only defined on the training/source samples of $x_i$).
- The approximation depends on the smoothness of $\pi$; dense couplings may blur the transport map.

**Question 1.** *Can we interpret coupling as a stochastic map where you can sample from the coupling? Does that also give us a way to get a map from the coupling?*

**Answer 1.** *Yes! This is exactly the probabilistic interpretation, where* $\tilde{T}(x) = \mathbb{E}_\pi[Y|X = x]$ *where we sample from* $\pi$.

**Question 2.** *What happens when we desire a map to the exact targets in the dataset rather than some linear combination of them?*

**Answer 2.** *In cases that we do not want interpolation, we can just choose the argmin or argmax over the distribution instead of taking the conditional expectation, but this is harder to compute.*

**Question 3.** *Is* $\tilde{T}$ *invertible?*

**Answer 3.** *No, it is not one-to-one and definitely not invertible. But you can take the transpose of the coupling to get a coupling from target to source (via the symmetry of the Kantorovich formulation) that can then be projected.*

**Question 4.** *Does this have a connection with clustering?*

**Answer 4.** *Yes. Each point of the source is mapping to the center/centroid of the target points it maps to (this is why it is referred to as barycentric). Think of this as a statistical centroid.*

**Question 5.** *Can we approximate new out-of-sample points using a nearest-neighbor approach?*

**Answer 5.** *Yes, we can find the nearest neighbor (the nearest train/source point) and then apply* $\tilde{T}$.

## 4.4 Entropic Transport Map

It turns out that in the case of *entropic optimal transport*, we *can* actually define a principled, smooth extension valid for any $x \in \mathcal{X}$!

Let $g_\varepsilon$ denote the optimal dual potential associated with the entropy-regularized OT problem. For any $x \in \mathcal{X}$, define the entropic transport map:

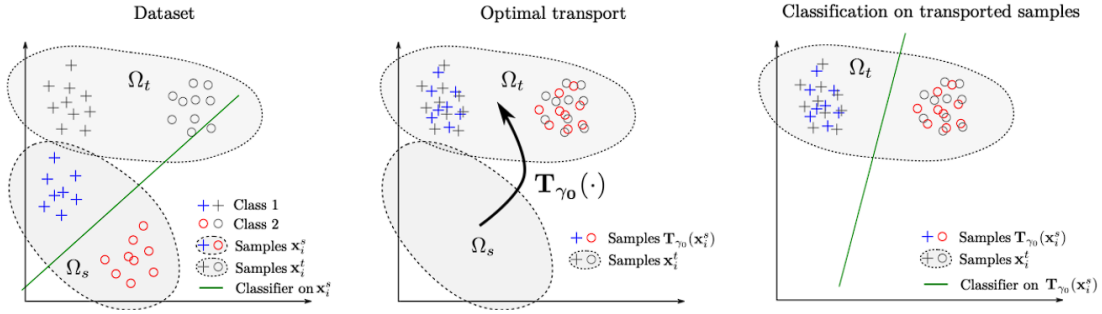
$$T_\varepsilon(x) = \frac{\int_y y \exp\left(\frac{1}{\varepsilon}\left(g_\varepsilon(y) - c(x,y)\right)\right) d\beta(y)}{\int_y \exp\left(\frac{1}{\varepsilon}\left(g_\varepsilon(y) - c(x,y)\right)\right) d\beta(y)}.$$

This map corresponds to the expectation of $y$ under the Gibbs distribution with energy given by the cost $c(x, y)$ and potential $g_\varepsilon(y)$.

- Pooladian & Niles-Weed (2021) show that $T_\varepsilon = \mathrm{Id} - \nabla f_\varepsilon$, where $f_\varepsilon$ is the optimal entropic dual potential.
- $T_\varepsilon(x)$ can be interpreted as the expected target location for $x$ under the Gibbs distribution defined by $(g_\varepsilon, c)$.
- One can think of $g_\varepsilon$ as the dual potential and $c$ as the cost potential.

Thus, in addition to efficient computing, entropy-regularized OT also provides a method for finding smooth approximate maps. This makes entropic OT especially powerful for applications requiring *out-of-sample generalization*, such as domain adaptation, generative modeling, and style transfer.

For example, suppose we want to do domain adaptation and have a classifier on some distribution $P = \Omega$, where the distribution drifted smoothly (so that now $Q = \Omega'$ is unlabeled). We push labeled data from $P$ into $Q$ to get labeled data for $Q$, and thus we can now train a classifier for $Q$.




**Question 6.** *Does this work if the transform is more rough or if the classes are permuted after the map?*

**Answer 6.** *This relies on the assumption that drift is smooth.*

# 5 Entropic Dual

## 5.1 Dual Problem

For entropy-regularized OT with marginals $a \in \Delta^n$, $b \in \Delta^m$, cost $C \in \mathbb{R}^{n\times m}$, and regularization parameter $\varepsilon > 0$, the dual reads:

$$\max_{f\in\mathbb{R}^n,\ g\in\mathbb{R}^m} \Big(\langle f, a\rangle + \langle g, b\rangle - \varepsilon\langle e^{f/\varepsilon}, Ke^{g/\varepsilon}\rangle\Big),$$

where

$$K_{ij} = e^{-C_{ij}/\varepsilon}.$$

Equivalently:

$$\max_{f,g} \langle f, \alpha\rangle + \langle g, \beta\rangle - \varepsilon \sum_{i,j} \exp\!\left(\tfrac{f_i+g_j-C_{ij}}{\varepsilon}\right).$$

## 5.2 Strong Duality and Dual Potentials

$$\text{Entropic primal value} \ \equiv\ \text{Entropic dual value.}$$

The optimal dual potentials $(f, g)$ are linked to Sinkhorn scalings $(u, v)$ by

$$u = e^{f/\varepsilon}, \qquad v = e^{g/\varepsilon}.$$

Combining with the primal solution form gives:

$$\pi^\star = \mathrm{Diag}(u)\, K\, \mathrm{Diag}(v),$$

or equivalently, entrywise:

$$\pi^\star_{ij} = \exp\!\left(\tfrac{f_i+g_j-C_{ij}}{\varepsilon}\right).$$

## 5.3 General Case

Generally, we have

$$\sup_{(f,g)\in C(X)\times C(Y)} \int f d\alpha + \int g d\beta - \varepsilon \int_{X\times Y} \exp\left(\frac{f(x)+g(y)-c(x,y)}{\varepsilon}\right) d\alpha(x) d\beta(y).$$

This corresponds to smoothing the constraint $f \oplus g \le c$. As $\varepsilon \to 0$, the general form recovers the non-entropic dual since the 3rd term becomes a hard constraint.

*Proof.* Define

$$\Phi(f, g) = \int f\, d\alpha + \int g\, d\beta - \varepsilon \iint e^{(f(x)+g(y)-c(x,y))/\varepsilon}\, d\alpha(x)\, d\beta(y).$$

Since $\Phi(f+\tau, g-\tau) = \Phi(f, g)$, we fix a gauge (e.g. $\int f\, d\alpha = 0$) and set $u = e^{f/\varepsilon},\ v = e^{g/\varepsilon} > 0$, giving

$$\Phi(f, g) = \varepsilon \int \log u\, d\alpha + \varepsilon \int \log v\, d\beta - \varepsilon \iint u(x)K(x, y)v(y)\, d\alpha d\beta, \quad K = e^{-c/\varepsilon}.$$

Maximizing sequences are bounded in $L^1$, so compactness and semicontinuity arguments yield existence of a maximizer $(u^\star, v^\star)$, hence $(f^\star, g^\star)$. As $\varepsilon \to 0$, the exponential penalty enforces $f(x) + g(y) \le c(x, y)$ almost everywhere; otherwise $\Phi(f, g) \to -\infty$. Thus the entropic dual converges to the classical Kantorovich dual

$$\sup_{f\oplus g\le c} \int f\, d\alpha + \int g\, d\beta.$$

□

# 6 Log-Domain

## 6.1 Dual Optimization via Block Ascent

The entropic dual objective is

$$H_\varepsilon(f,g) \;=\; \langle f, a\rangle + \langle g, b\rangle - \varepsilon\,\langle e^{f/\varepsilon},\, Ke^{g/\varepsilon}\rangle,$$

where $K = \exp(-C/\varepsilon)$ and $(f,g)$ are dual potentials. A natural optimization strategy is block coordinate ascent in $(f,g)$.

The gradients of $H_\varepsilon$ are:

$$\nabla_f H_\varepsilon(f,g) = a - e^{f/\varepsilon} \odot (Ke^{g/\varepsilon}), \qquad \nabla_g H_\varepsilon(f,g) = b - e^{g/\varepsilon} \odot (K^\top e^{f/\varepsilon}).$$

Setting each gradient to zero and solving in alternation yields:

$$f^{(\ell+1)} = \varepsilon \log a \;-\; \varepsilon \log\!\big(Ke^{g^{(\ell)}/\varepsilon}\big),$$

$$g^{(\ell+1)} = \varepsilon \log b \;-\; \varepsilon \log\!\big(K^\top e^{f^{(\ell+1)}/\varepsilon}\big).$$

Define the scaling vectors

$$u = e^{f/\varepsilon}, \qquad v = e^{g/\varepsilon}.$$

Then the updates become

$$u^{(\ell+1)} = a \oslash (Kv^{(\ell)}), \qquad v^{(\ell+1)} = b \oslash (K^\top u^{(\ell+1)}),$$

which are exactly the iterations of the Sinkhorn algorithm. The log-domain formulation shows:

- Sinkhorn iterations correspond to block coordinate ascent on the entropic dual.
- Potentials $(f,g)$ and scalings $(u,v)$ are related by $u = e^{f/\varepsilon}$, $v = e^{g/\varepsilon}$.
- The log-domain view avoids numerical underflow and clarifies the variational structure of Sinkhorn.

## 6.2 Log-domain Sinkhorn

Starting from the entropy-regularized OT coupling

$$\pi_{ij} = u_i K_{ij} v_j, \qquad K_{ij} = \exp(-C_{ij}/\varepsilon),$$

with updates

$$u \leftarrow \frac{a}{Kv}, \qquad v \leftarrow \frac{b}{K^\top u},$$

define log-scalings

$$f = \varepsilon \log u, \qquad g = \varepsilon \log v.$$

Then

$$\pi_{ij} = \exp\!\left(\tfrac{f_i + g_j - C_{ij}}{\varepsilon}\right).$$

The log-updates follow from the marginal constraints:

$$f_i = \varepsilon \log a_i + \mathrm{softmin}_\varepsilon\big(C_{i\cdot} - g\big), \quad g_j = \varepsilon \log b_j + \mathrm{softmin}_\varepsilon\big(C_{\cdot j} - f\big),$$

where

$$\mathrm{softmin}_\varepsilon(z) := -\varepsilon \log \sum_k e^{-z_k/\varepsilon}.$$

**Stable iteration:**

$$f \leftarrow \varepsilon \log a + \mathrm{softmin}_\varepsilon(C - \mathbf{1}g^\top)$$
$$g \leftarrow \varepsilon \log b + \mathrm{softmin}_\varepsilon(C^\top - \mathbf{1}f^\top).$$

### 6.3 Limit Behavior and Stability

As $\varepsilon \to 0$, these conditions enforce

$$f \oplus g \;\le\; C, \qquad f_i \le \min_j (C_{ij} - g_j),$$

corresponding to the hard inequality constraints of the non-entropic dual. However, naive enforcement without the log-sum-exp relaxation does not yield convergence.

Direct computation of $\min_\varepsilon(z)$ can be unstable if $\{z_i\}$ vary widely. To stabilize, we use the log-sum-exp trick:

$$\min_\varepsilon(z) = m - \varepsilon \log \sum_{i=1}^{n} e^{-(z_i - m)/\varepsilon}, \qquad m = \min_i z_i,$$

which avoids overflow/underflow. This is the standard LSE implementation in frameworks such as PyTorch and NumPy.

## 7 EOT vs OT

### 7.1 Limiting Behavior and Bounds

Let $\mathrm{OT}_\varepsilon$ denote the entropy-regularized OT cost. As $\varepsilon \to 0$,

$$\mathrm{OT}_\varepsilon \;\longrightarrow\; \mathrm{OT},$$

i.e. the entropic formulation converges to the classical OT cost.

We also have the sandwich inequality:

$$\mathrm{OT} - \varepsilon H(\pi_\varepsilon^\star) \;\le\; \mathrm{OT}_\varepsilon \;\le\; \mathrm{OT},$$

where $\pi_\varepsilon^\star$ is the entropic optimal coupling.

**Proof of Upper Bound.** Evaluate the entropic objective at any unregularized OT optimizer $\pi^\star$:

$$\mathrm{OT}_\varepsilon \;\le\; \langle C, \pi^\star \rangle - \varepsilon H(\pi^\star) \;\le\; \langle C, \pi^\star \rangle = \mathrm{OT},$$

since $H(\pi^\star) \ge 0$.

**Proof of Lower Bound.** By optimality of $\pi_\varepsilon^\star$:

$$\mathrm{OT}_\varepsilon = \langle C, \pi_\varepsilon^\star \rangle - \varepsilon H(\pi_\varepsilon^\star) \;\ge\; \min_{\pi \in \Pi(\alpha,\beta)} \Big( \langle C, \pi \rangle - \varepsilon H(\pi) \Big).$$

In particular,

$$\mathrm{OT}_\varepsilon \;\ge\; \mathrm{OT} - \varepsilon H(\pi_\varepsilon^\star).$$

**Theorem 2.** *We can find a lower bound that does not depend explicitly on $\pi_\varepsilon^\star$.*

*Proof.* Start from the primal definition

$$\mathrm{OT}_\varepsilon = \min_{\pi \in \Pi(\alpha,\beta)} \Big\{ \langle C, \pi \rangle - \varepsilon H(\pi) \Big\} \;\ge\; \min_{\pi \in \Pi(\alpha,\beta)} \langle C, \pi \rangle \;-\; \varepsilon \sup_{\pi \in \Pi(\alpha,\beta)} H(\pi).$$

Thus it suffices to upper bound the entropy over all feasible couplings. Using the KL identity with the independent coupling $\alpha \otimes \beta$,

$$H(\pi) \;=\; H(\alpha \otimes \beta) - \mathrm{KL}\big(\pi \,\|\, \alpha \otimes \beta\big) \;\le\; H(\alpha \otimes \beta) \;=\; H(\alpha) + H(\beta),$$

since $\mathrm{KL} \ge 0$ and $H(\alpha \otimes \beta) = H(\alpha) + H(\beta)$. Therefore

$$\boxed{\mathrm{OT}_\varepsilon \;\ge\; \mathrm{OT} \;-\; \varepsilon\big(H(\alpha) + H(\beta)\big).}$$

- This bound is sharp when the maximizing coupling is the product $\alpha \otimes \beta$ (e.g., for $C \equiv \text{const}$).
- A looser but assumption-free alternative is $H(\pi) \leq \log(nm)$ on an $n \times m$ grid, giving $\text{OT}_\varepsilon \geq \text{OT} - \varepsilon \log(nm)$.

□

## 7.2 Sinkhorn Divergence

A key issue with Entropic Optimal Transport (EOT) is that it does not yield a true metric:

$$W_\varepsilon(\alpha, \alpha) > 0,$$

i.e., a probability distribution has nonzero cost to itself.

- This means EOT is **biased** as a dissimilarity measure.
- Consequently, $W_\varepsilon$ cannot be directly used as a "distance" in applications (e.g., in ML tasks, domain adaptation, or generative modeling).

To remove this bias, the **Sinkhorn Divergence** (SD) proposes:

$$S_\varepsilon(\alpha, \beta) \;=\; OT_\varepsilon(\alpha, \beta) - \tfrac{1}{2} OT_\varepsilon(\alpha, \alpha) - \tfrac{1}{2} OT_\varepsilon(\beta, \beta).$$

- This subtraction debiases the entropic OT cost, ensuring $S_\varepsilon(\alpha, \alpha) = 0$.
- The correction terms remove the self-similarity bias inherent in EOT.

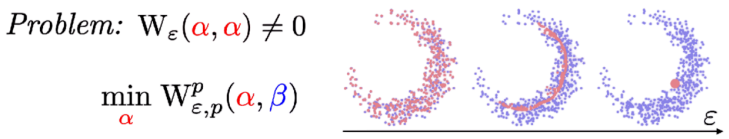
The Sinkhorn Divergence has several desirable properties:

- **Symmetric**: $S_\varepsilon(\alpha, \beta) = S_\varepsilon(\beta, \alpha)$.
- **Nonnegative and definite**: $S_\varepsilon(\alpha, \beta) \geq 0$, and $S_\varepsilon(\alpha, \beta) = 0 \iff \alpha = \beta$.
- **Interpolates between metrics**:

$$\lim_{\varepsilon \to \infty} S_\varepsilon(\alpha, \beta) \quad \longrightarrow \quad \text{MMD (Maximum Mean Discrepancy)},$$

$$\lim_{\varepsilon \to 0} S_\varepsilon(\alpha, \beta) \quad \longrightarrow \quad OT(\alpha, \beta).$$

  Thus, $S_\varepsilon$ provides a smooth family of divergences bridging kernel methods and OT.
- **Metric-like and stable**: In practice, $S_\varepsilon$ behaves like a metric, and is widely used in ML applications where computational stability is important.




There are examples/papers on Learning generative models with Sinkhorn divergence [example].

## 7.3 Sample Complexity

The **sample complexity** of OT refers to the difficulty of estimating the true transport cost from finite samples. Given $n$ samples from distributions $\alpha$ and $\beta$, we consider their empirical measures $\hat{\alpha}_n$, $\hat{\beta}_n$, and study:

$$W_p(\hat{\alpha}_n, \hat{\beta}_n) \quad \longrightarrow \quad W_p(\alpha, \beta).$$

Unfortunately, estimating OT is statistically hard:

$$\mathbb{E}\Big[\,\big|W_p(\hat{\alpha}_n,\hat{\beta}_n) - W_p(\alpha,\beta)\big|\,\Big] = O\Big(n^{-1/d}\Big).$$

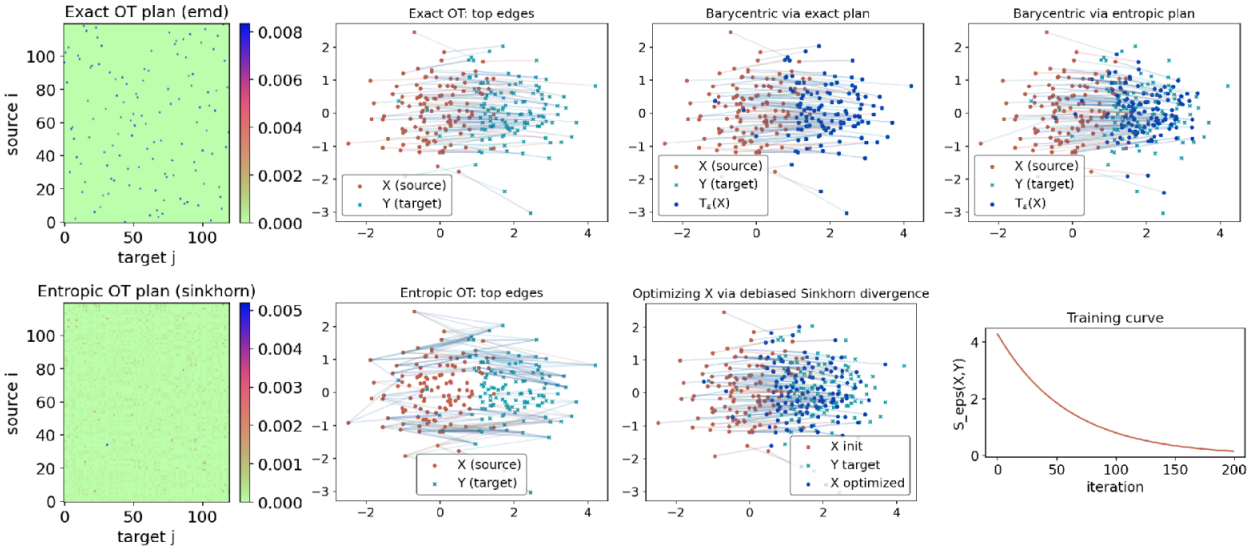
This **slow convergence rate** is known as the curse of dimensionality:

- Convergence worsens dramatically with higher ambient dimension $d$.
- Makes OT estimation infeasible in many machine learning contexts with high-dimensional data.

With entropic regularization, the statistical picture improves considerably:

$$\mathbb{E}\Big[\,\big|W_{\varepsilon,p}(\hat{\alpha}_n,\hat{\beta}_n) - W_{\varepsilon,p}(\alpha,\beta)\big|\,\Big] = O\Big(\varepsilon^{-\frac{d}{2}} n^{-1/2}\Big).$$

- For fixed $\varepsilon$, the rate is $O(n^{-1/2})$, the **parametric rate**, independent of dimension $d$.
- As $\varepsilon \to \infty$, the bound converges exactly to $O(n^{-1/2})$, the best achievable statistical rate.
- Empirically, entropic OT estimators are much more stable in high-dimensional ML applications.




# References


[PC19] Gabriel Peyré and Marco Cuturi. Computational optimal transport. *Foundations and Trends in Machine Learning*, 11(5-6):355–607, 2019. (See also the arXiv preprint arXiv:1803.00567).